

Experiment 12: Port Scanning Tools

Aim

To explore and practice different types of port scanning using Nmap and other tools available in Kali Linux.

Procedure

1. **TCP SYN Scan (Stealth / Half-open scan):**
`nmap -sS scanme.nmap.org`
2. **TCP Connect Scan:**
`nmap -sT scanme.nmap.org`
3. **UDP Scan:**
`nmap -sU -p 53,161 scanme.nmap.org`
4. **Scan specific ports:**
`nmap -p 22,80,443 scanme.nmap.org`
5. **Scan a range of ports:**
`nmap -p 1-1000 scanme.nmap.org`

Sample Output / Results

```
$ nmap -sS -p 22,80,443 scanme.nmap.org
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 81213sec preferred_lft 81213sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86222sec preferred_lft 14222sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)~  
$ nmap scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 12:02 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.022s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 987 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    closed domain  
80/tcp    open  http  
83/tcp    closed mit-ml-dev  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
993/tcp   closed imaps  
2022/tcp  closed down  
3389/tcp  closed ms-wbt-server  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
8080/tcp  closed http-proxy  
Nmap done: 1 IP address (1 host up) scanned in 20.28 seconds  
(kali@kali)~  
$ nmap -p- 192.168.1.10  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 12:30 -0400  
Nmap scan report for 192.168.1.10  
Host is up (0.026s latency).  
All 65535 scanned ports on 192.168.1.10 are in ignored states.  
Not shown: 46375 filtered tcp ports (no-response), 19160 filtered tcp ports (net-unreach)  
Nmap done: 1 IP address (1 host up) scanned in 204.53 seconds
```